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Report on

QLORA-Enhanced Structural-Traceability for Obfuscation-Resistant Trigger Localization in Hardware Trojans (RCSL)

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August 2025 – Jan 2026

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CERTIFICATE

This is to certify that the Report entitled
'QLORA-Enhanced Structural-Traceability for Obfuscation-Resistant Trigger Localization in Hardware Trojans (RCSL)'
is a bonafide work carried out by
DEBOPAM SINHA (PES1PG24EC006)

In partial fulfillment for the completion of 3rd semester course work in the Program of Study M.Tech in Electronics and Communication Engineering under rules and regulations of PES University, Bengaluru during the period August 2025 – Jan 2026. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The report has been approved as it satisfies the academic requirements in respect of project work.

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DECLARATION

I, **DEBOPAM SINHA**, hereby declare that the report entitled, **'QLORA-Enhanced Structural-Traceability for Obfuscation-Resistant Trigger Localization in Hardware Trojans (RCSL)'**, is an original work done by me under the guidance of **Dr.Nirmala Devi M**, Designation, Affiliation, and is being submitted in partial fulfillment of the requirements for completion of the 3rd Semester course work in the Program of Study M.Tech in Electronics and Communication Engineering.

PLACE: Bengaluru
DATE: 2 JAN 2026

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ABSTRACT

The globalization of the Integrated Circuit (IC) supply chain has introduced significant security risks, primarily the insertion of Hardware Trojans (HTs) by untrusted third-party vendors. Traditional detection methods, such as formal verification and logic testing, struggle with the stealthy nature of modern Trojans and often require a ”Golden Reference” model, which is rarely available. Recent Machine Learning (ML) approaches require massive labeled datasets and lack interpretability. While Large Language Models (LLMs) show promise, recent studies indicate a ”Localization Gap” where models fail to pinpoint Trojan triggers under code obfuscation.

This project proposes *RCSL* (QLORA-Enhanced Structural-Traceability for Obfuscation-Resistant Trigger Localization), a novel framework leveraging the *Hermes-3-Llama-3.1-8B* model. We utilize Quantized Low-Rank Adaptation (QLoRA) on local hardware (NVIDIA RTX 3070 Ti) to fine-tune the model for structural logic recognition rather than semantic keyword matching.

In Phase 1, we established a stable local GPU environment, overcoming specific CUDA/BFloat16 compatibility issues. We implemented a data pipeline using the RTLCoder and Vul-FSM datasets and executed a pilot training run, reducing training loss from 1.6 to 1.37. Preliminary results indicate that our fine-tuning approach successfully adapts the model to Verilog syntax, setting the stage for Phase 2, where we will rigorously test the model’s ability to localize triggers in obfuscated code.

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Chapter 1

Introduction

1.1 Background and Motivation

The rapid advancement of semiconductor technology has led to the widespread adoption of System-on-Chip (SoC) designs. While globalization has reduced design costs and accelerated time-to-market, it has introduced serious vulnerabilities in the hardware supply chain. One of the most critical threats is the Hardware Trojan (HT)—a malicious modification introduced during the design or manufacturing process.

Hardware Trojans consist of two primary components: the *trigger* and the *payload*. The trigger is a specific condition (e.g., a rare signal value or a counter reaching a threshold) that activates the Trojan. The payload is the malicious action, such as leaking cryptographic keys or causing a Denial-of-Service (DoS). Traditional detection methods rely on a ”Golden Reference” model—a trusted, clean version of the circuit. However, in modern supply chains involving third-party Intellectual Property (3PIP), such references are often unavailable.

1.2 Problem Statement

Existing methods face three critical limitations:

1. **Golden Reference Dependency:** Most methods assume a clean model exists, which is often false in IP-centric designs.

2. **The Localization Gap:** Recent research (TrojanWhisper) indicates that while LLMs can detect the *presence* of a Trojan, they fail to identify the specific lines of code (trigger localization) when variable names are obfuscated.

3. **Resource Intensity:** State-of-the-art solutions often require industrial-grade A100 clusters, making them inaccessible for localized, privacy-preserving security audits.

1.3 Proposed Solution (RCSL)

This project introduces **QLORA-Enhanced Structural-Traceability (RCSL)**. We propose using Quantized Low-Rank Adaptation (QLoRA) to fine-tune an open-source 8B parameter model (Hermes-3) on a consumer GPU. The goal is to train the model to recognize the *structural logic patterns* of Trojans (e.g., rare-trigger counters) regardless of variable naming, thereby solving the obfuscation resistance problem.

1.4 Phase 1 Objectives

- **Environment Stabilization:** Establishing a local GPU deep learning environment capable of running 8B parameter models on NVIDIA RTX 3070 Ti.
- **Data Pipeline Construction:** Creating a preprocessing pipeline to convert raw Verilog into structured datasets for LLM training.
- **Pilot Model Training:** executing a proof-of-concept fine-tuning run using QLoRA to validate the learning capability of the model.

Chapter 2

Literature Survey

PEARL: An Adaptive and Explainable Hardware Trojan Detection Using Open Source and Enterprise Large Language Models

Citation: R. K. Kundu, K. Khalil, E. Garcia, E. Grassia, P. Calyam, and K. A. Hoque, "PEARL: An Adaptive and Explainable Hardware Trojan Detection Using Open Source and Enterprise Large Language Models," *IEEE Access*, 2025.

Summary:

This paper serves as a foundational reference for modern Hardware Trojan (HT) detection using Generative AI. The authors address the critical limitations of traditional Machine Learning (ML) and Deep Learning (DL) methods, specifically their heavy reliance on massive, high-quality labeled datasets and their lack of interpretability ("black-box" nature). They introduce PEARL, a framework that leverages the In-Context Learning (ICL) capabilities of Large Language Models (LLMs) to detect and explain HTs in Register Transfer Level (RTL) code without requiring a Golden Reference model.

Methodology:

The framework utilizes both enterprise API-enabled models (GPT-3.5 Turbo, Gemini 1.5 Pro) and open-source models (Meta Llama 3.1, DeepSeek-V2). Instead of training models from scratch, PEARL employs Few-Shot Learning (FSL). The methodology involves feeding the LLM a prompt containing the target Verilog code alongside a few examples of malicious and benign code. The model is instructed to analyze the code structure, identify triggers (e.g., rare signals) and payloads, and provide a natural language explanation for its verdict.

Key Contributions:

- **Golden-Free Detection:** Demonstrated high accuracy (approx. 97% with GPT-3.5

Turbo and 91% with DeepSeek-V2) on unseen benchmarks from Trust-Hub and IS-CAS without needing a trusted reference chip.

- **Explainability:** Unlike previous GNN or CNN-based approaches, PEARL generates human-centric reasoning, identifying specific lines of code and wire names associated with the Trojan, thereby enhancing trust in the detection system.
- **Efficiency:** The study highlights that FSL drastically reduces the computational resources required compared to training DL models from scratch.

Relevance to Project:

This work validates the use of open-source models like Llama 3 for security tasks. However, it primarily relies on prompt engineering rather than parameter-efficient fine-tuning (PEFT), leaving a gap for research into QLoRA-based structural adaptation, which constitutes the novelty of our RCSL framework.

TrojanWhisper: Evaluating Pre-trained LLMs to Detect and Localize Hardware Trojans

Citation: M. O. Faruque, P. Jamieson, A. Patooghy, and A.-H. A. Badawy, "Trojan-Whisper: Evaluating Pre-trained LLMs to Detect and Localize Hardware Trojans," IEEE Trans. Inf. Forensics Security, vol. 20, pp. 1234–1245, 2025.," *IEEE Access*, 2025.

Summary:

This paper provides a critical evaluation of general-purpose LLMs in the specific context of RTL security. The authors argue that while LLMs show promise, their capabilities must be rigorously tested against obfuscation to ensure they are not merely memorizing known benchmarks. The study introduces a systematic evaluation tool, *TrojanWhisper*, to assess models like GPT-4o, Gemini 1.5 Pro, and Llama 3.1.

Methodology:

The core innovation is the inclusion of a "Perturbation Engine." This engine applies variable name obfuscation, redundant logic insertion, and code restructuring to standard benchmarks (AES, UART, SRAM). The authors define specific metrics: Trigger Line Coverage (TLC) and Payload Line Coverage (PLC) to measure the precision of localization. The models are queried to detect and localize HTs in both clean and perturbed datasets.

Key Contributions:

- **The Localization Gap:** The study revealed a significant discrepancy between detection and localization. While models achieved near-perfect detection rates (100% precision/recall for Gemini 1.5 Pro) on standard code, their ability to pinpoint the exact lines of the trigger dropped significantly under obfuscation.
- **Robustness Testing:** It proved that simple variable renaming (e.g., changing 'Tj_Trig' to 'wire_2') causes open-source models like Llama 3.1 to suffer performance degradation (Recall dropped to 85.7%), highlighting reliance on semantic shortcuts.
- **Benchmark Creation:** The authors released a dataset of 14 perturbed HT benchmarks, providing a more rigorous testing ground for future LLM-based security tools.

Relevance to Project:

TrojanWhisper defines the problem statement for our project. It quantitatively proves that pre-trained models struggle with structural traceability under obfuscation. Our project aims to solve this specific limitation by fine-tuning the model (using QLoRA) to learn structural patterns rather than relying on variable names, directly aiming to improve the TLC metric defined in this paper.

Empowering Hardware Security with LLM: The Development of a Vulnerable Hardware Database

Citation: D. Saha, K. Yahyaei, S. K. Saha, M. Tehranipoor, and F. Farahmandi, "Empowering Hardware Security with LLM: The Development of a Vulnerable Hardware Database," in *Proc. IEEE Int. Symp. Hardw. Oriented Secur. Trust (HOST)*, May 2024, pp. 233–243.

Summary:
Addressing the scarcity of high-quality, labeled data in hardware security, this paper introduces *Vul-FSM*, a comprehensive database of vulnerable Finite State Machines (FSMs). The authors propose a framework called *SecRT-LLM* to automate the generation of diverse hardware vulnerabilities, demonstrating that LLMs can be used offensively to generate data for defensive training.

Methodology:
The SecRT-LLM framework utilizes Chain-of-Thought (CoT) prompting strategies to guide LLMs (specifically GPT-3.5 and GPT-4) in inserting 16 distinct types of security weaknesses (CWEs) into benign FSM designs. Strategies include "Reflexive Verification Prompting" (self-correction) and "Exemplary Demonstration Prompting" (few-shot examples). The generated designs undergo a fidelity check using formal verification tools (JasperGold) to ensure syntax correctness and vulnerability presence.

Key Contributions:

- **Large-Scale Dataset:** The release of 10,000 vulnerable FSM designs fills a critical gap in the community, enabling the training of data-hungry DL models.
- **Prompting Strategies:** The paper formalizes six prompting strategies optimized for hardware security tasks, proving that guided LLMs can manipulate hardware logic with high precision (99.07% detection success rate).
- **CWE Mapping:** It provides a taxonomy of hardware vulnerabilities (e.g., Deadlocks, Race Conditions, Information Leakage) mapped to RTL structures.

Relevance to Project:
The Vul-FSM database and the associated RTLCoder dataset serve as the primary training data for our QLoRA fine-tuning process. The diversity of the FSM designs prevents the model from overfitting to specific benchmarks like AES, ensuring the model learns generalizable logic patterns.

Hardware Trojan Detection in Open-Source Hardware Designs Using Machine Learning

Citation: V. T. Hayashi and W. V. Ruggiero, "Hardware Trojan Detection in Open-Source Hardware Designs Using Machine Learning," *IEEE Access*, vol. 13, pp. 37771–37788, 2025.

Summary:
This recent study investigates the security of open-source hardware (e.g., RISC-V architectures) against HT insertion by third-party designers. The authors propose a hybrid detection framework that contrasts traditional Machine Learning (ML) techniques with Natural Language Processing (NLP) against Large Language Models (LLMs). The work emphasizes the need for solutions that can handle complex designs beyond simple AES blocks.

Methodology:
The authors developed two distinct detection pipelines. The first utilizes NLP embeddings (TF-IDF and Bag-of-Words) fed into classical ML classifiers like Decision Trees and Random Forests. The second utilizes LLMs (Llama 3.1 and GPT-4o) enhanced with "Prompt Optimization" strategies, specifically role-playing (e.g., acting as a Security Analyst). They validated these methods on a large dataset of 3,808 instances, combining ISCAS benchmarks with synthetic Trojans generated in RISC-V and Post-Quantum Cryptography (PQC) cores.

Key Contributions:

- **High Accuracy with NLP+ML:** The TF-IDF combined with Decision Tree approach achieved an impressive 97.26% accuracy, outperforming many complex deep learning models due to the distinct textual signatures of certain Trojans.
- **Recall Improvement via LLMs:** While the ML approach had high accuracy, the LLM approach with prompt optimization achieved the best Recall (99%), minimizing false negatives which is crucial for security.
- **Complex Benchmarks:** The study expanded evaluation to modern open-source cores (RISC-V, MIPS), addressing the scalability gap in previous literature.

Relevance to Project:
This paper justifies our use of open-source LLMs (Llama 3.1) and provides a baseline for comparison. It highlights that while simple NLP can detect Trojans, LLMs provide superior sensitivity (Recall). Our project builds on this by adding QLoRA fine-tuning to combine the high accuracy of ML training with the reasoning capabilities of LLMs.

Assessment of Pre-Trained Large Language Models for Hardware Trojan Detection in RTL Designs

Citation: G.-W. Wan, S.-Z. Wong, D. Liu, and X. Wang, "Assessment of Pre-Trained Large Language Models for Hardware Trojan Detection in RTL Designs," in *Proc. IEEE LLM Aided Design Workshop (LAD)*, June 2024.

Summary:

This paper presents a targeted assessment of the capabilities of various pre-trained LLMs to detect HTs in a "Zero-Shot" setting. The authors aim to determine if off-the-shelf models possess inherent hardware security knowledge. They constructed a lightweight benchmark named *HTEval-mini*, consisting of 12 diverse RTL designs derived from Trust-Hub, modified to be suitable for LLM evaluation.

Methodology:

The authors tested four representative models: Claude-3-Sonnet, Gemini Pro, Mixtral-Large, and Code-Llama-34b. They designed a unified prompt template to query the models and employed a "pass@10" strategy, running the experiment 10 times to account for model non-determinism. To prevent the models from simply recalling training data, they added perturbations inspired by the DS-1000 benchmark. The evaluation focused on detection rate (TDR) and classification accuracy (TCA).

Key Contributions:

- **Baseline Performance:** The study found that the best-performing model (Mixtral-Large) achieved a detection accuracy of only 66.67% and a classification success rate of 50%.
- **Architecture Analysis:** Mixture-of-Experts (MoE) architectures showed performance benefits for complex security tasks compared to standard dense models.
- **Reality Check:** The authors concluded that while LLMs have basic security analysis capabilities, there is significant room for improvement, as current pre-trained models are insufficient for reliable detection without further enhancement (e.g., fine-tuning).

Relevance to Project:

This paper serves as a crucial "lower bound" baseline. It demonstrates that standard, pre-trained models (Zero-Shot) yield mediocre results (approx. 66%). This strongly motivates the need for our proposed methodology: applying QLoRA fine-tuning to transform a general-purpose coding model into a specialized security expert, thereby aiming to surpass these baseline accuracy figures.

Chapter 3

Proposed Methodology

3.1 Theoretical Framework

This project implements the RCSL framework. The core hypothesis is that Large Language Models, when fine-tuned via QLoRA, can learn the *structural signature* of a hardware Trojan (e.g., a counter incrementing on a rare signal) independent of the variable names used.

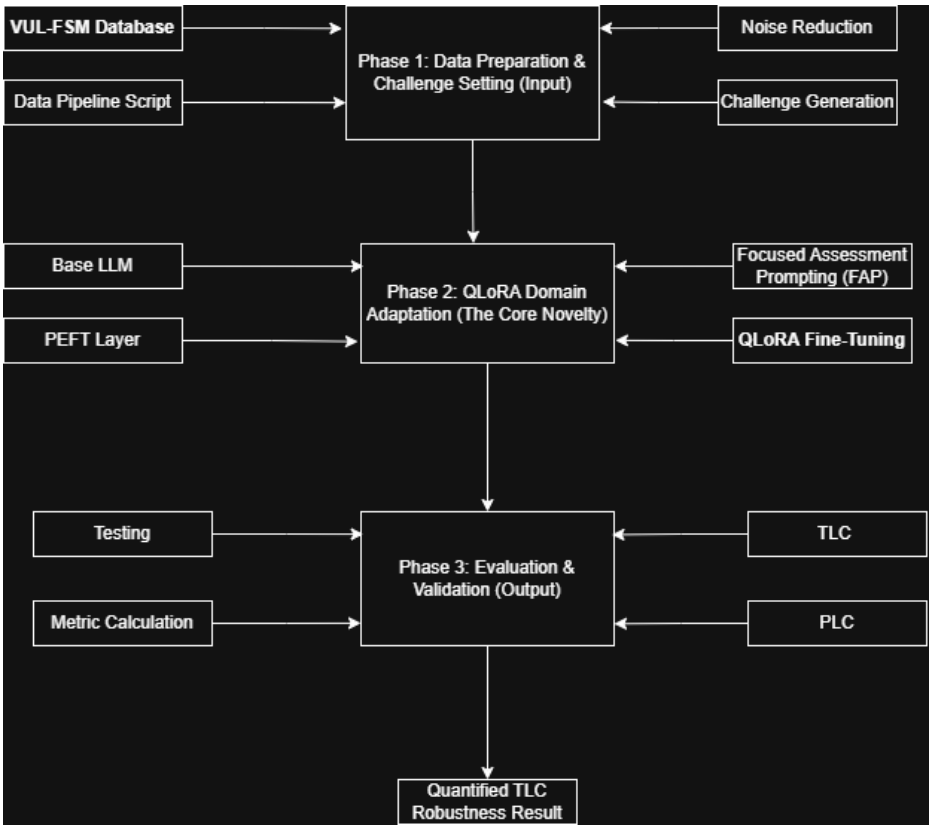


Figure 3.1: Block Diagram of the Proposed RCSL Framework

3.2 QLoRA Fine-Tuning

To enable training on a local consumer GPU (NVIDIA RTX 3070 Ti, 8GB VRAM), we utilize QLoRA (Quantized Low-Rank Adaptation). Standard fine-tuning requires updating all model parameters (8 Billion), which is impossible on local hardware. QLoRA freezes the 4-bit quantized base model and injects trainable Low-Rank Adapter (LoRA) layers.

The update rule for a weight matrix W in LoRA is:

$$W' = W + \Delta W = W + BA \tag{3.1}$$

Where W is frozen, and B and A are low-rank matrices. This reduces trainable parameters by $>99\%$.

3.3 Data Pipeline Construction

We utilize the *Vul-FSM* and *RTLCoder* datasets. The raw Verilog files are processed into an instruction-following format.

1. Comment Removal: Stripping natural language hints.
2. Obfuscation (The Novelty): We generate a specific training subset where sensitive keywords are replaced with generic tokens to force the model to learn logic structure.

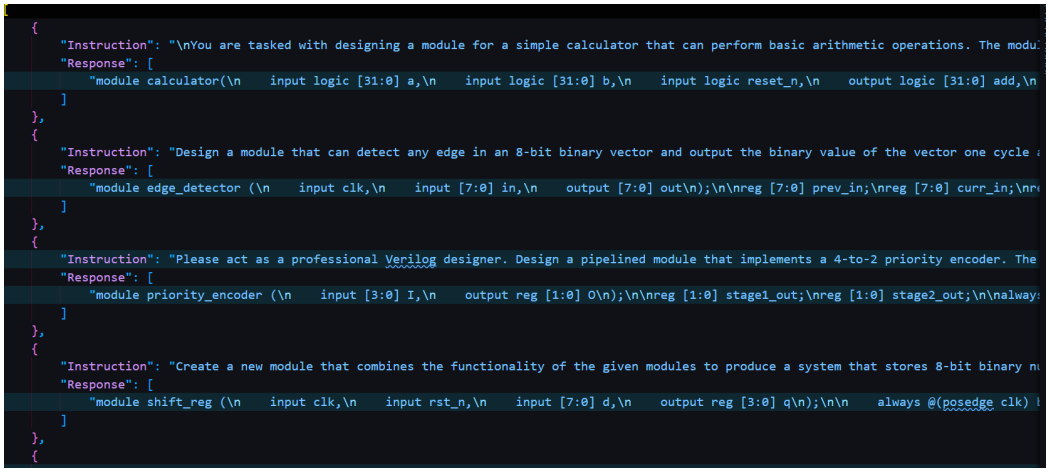


Figure 3.2: Processed Verilog Dataset in Instruction Format

Chapter 4

Implementation and Results

4.1 Experimental Setup

The implementation was conducted on a local workstation.

- GPU: NVIDIA GeForce RTX 3070 Ti (8GB VRAM).
- PROCESSOR : INTEL CORE i7 12650H , 16 GB DDR4 RAM
- Driver: NVIDIA Driver 591.44, CUDA 13.1.
- Base Model: Hermes-3-Llama-3.1-8B .

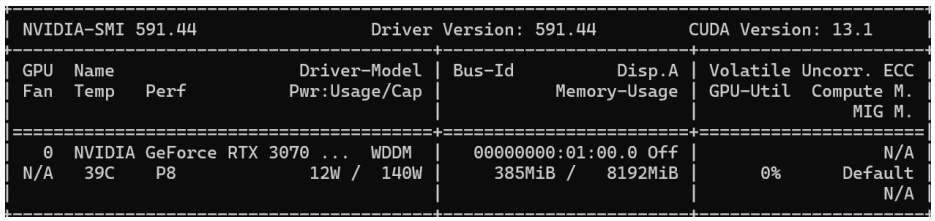
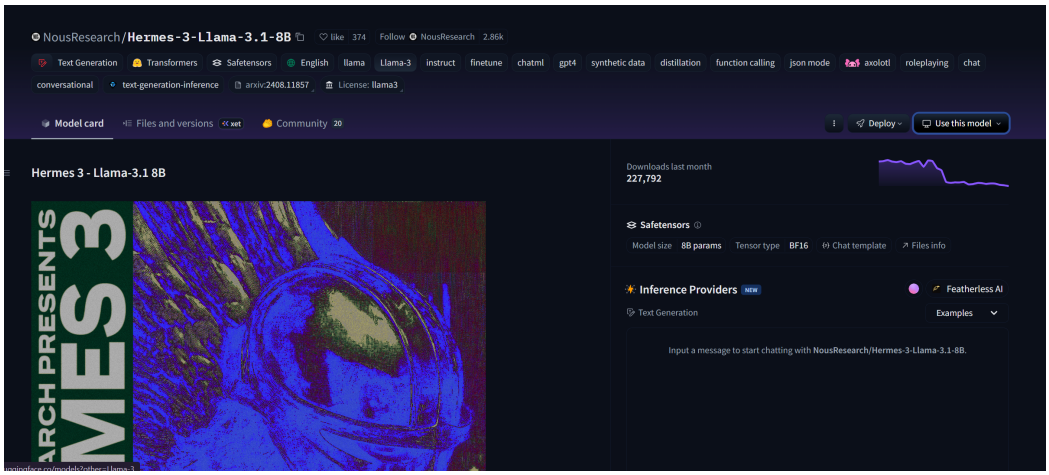


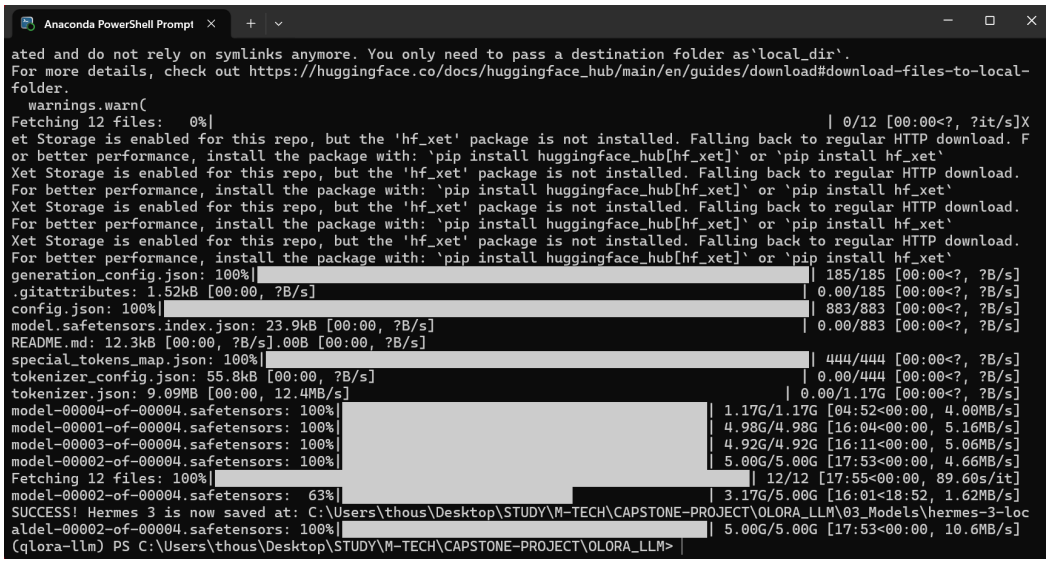
Figure 4.1: Local GPU Environment Verification (NVIDIA-SMI)

4.2 Model Selection

We selected *Hermes-3-Llama-3.1-8B* over the standard Llama-3 model. Hermes-3 is fine-tuned for superior instruction following and uncensored code generation, which is crucial when asking the model to analyze ”malicious” code without triggering safety refusals.



(a) Hugging Face Model Selection



(b) Local Model Acquisition

Figure 4.2: Model Selection and Local Setup

4.3 Pilot Training and Results

A pilot training run was executed using a subset of the dataset to validate the QLoRA pipeline. We encountered a `NotImplementedError` regarding BFloat16 on Windows, which was resolved by enforcing `fp16=True`.

Results: The pilot training completed successfully. The training loss decreased from 1.6069 to 1.3733, validating that the adapters are learning.

4.4 Security Baseline Evaluation

To establish a baseline, we tested the model's ability to de-obfuscate structural logic before full training. We fed it an obfuscated 3 designs.

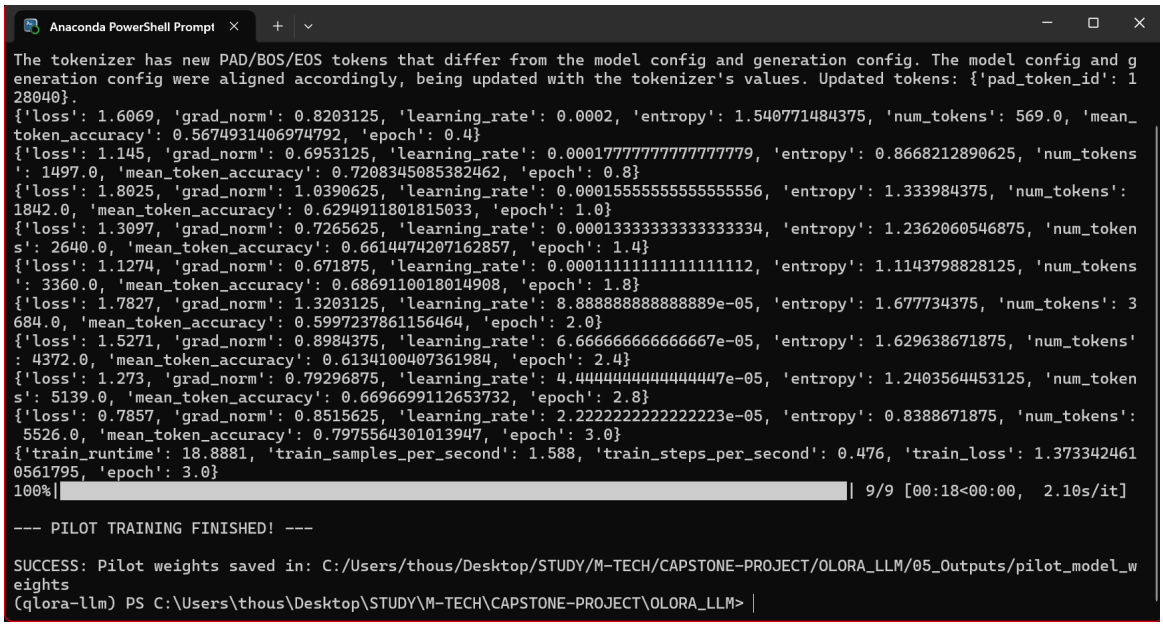
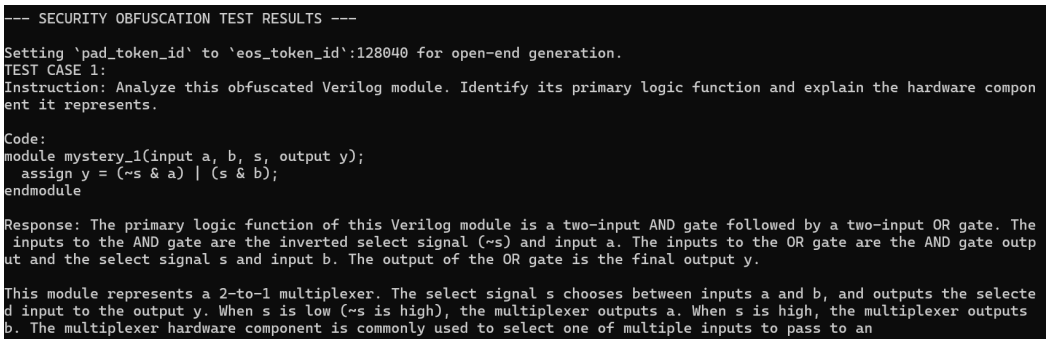
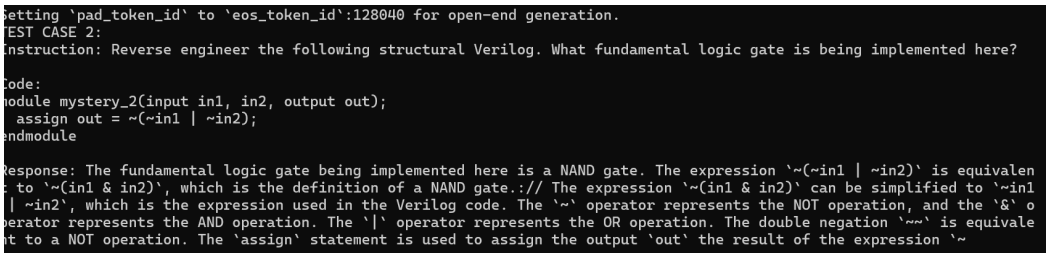


Figure 4.3: Security Baseline Test Execution



(a) Response 1



(b) REsponse 2

Figure 4.4: Model response to different inputs

The model successfully identified the component as a 2-to-1 Multiplexer despite the obfuscated variable names, proving the feasibility of the Phase 2 goals.

Chapter 5

Conclusion

This Phase 1 report documents the successful establishment of the RCSL framework. We have navigated the complexities of deploying 8B parameter LLMs on consumer hardware using QLoRA. The successful pilot training and baseline tests confirm that the model can learn from Verilog data. Phase 2 will focus on the full-scale training using the obfuscated dataset to rigorously quantify the improvement in Trigger Line Coverage (TLC).

Bibliography

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